

SIMATS ENGINEERING
CO4-AT2-Laboratory Performance

COURSE CODE: CSA0904

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COURSE NAME: Programming in JAVA

REG NO: 192421410

Q1 Basic Shapes

25 marks

Develop an Applet that draws basic shapes (line, rectangle, oval) and changes their color based on user selection from a drop-down list (Choice control).

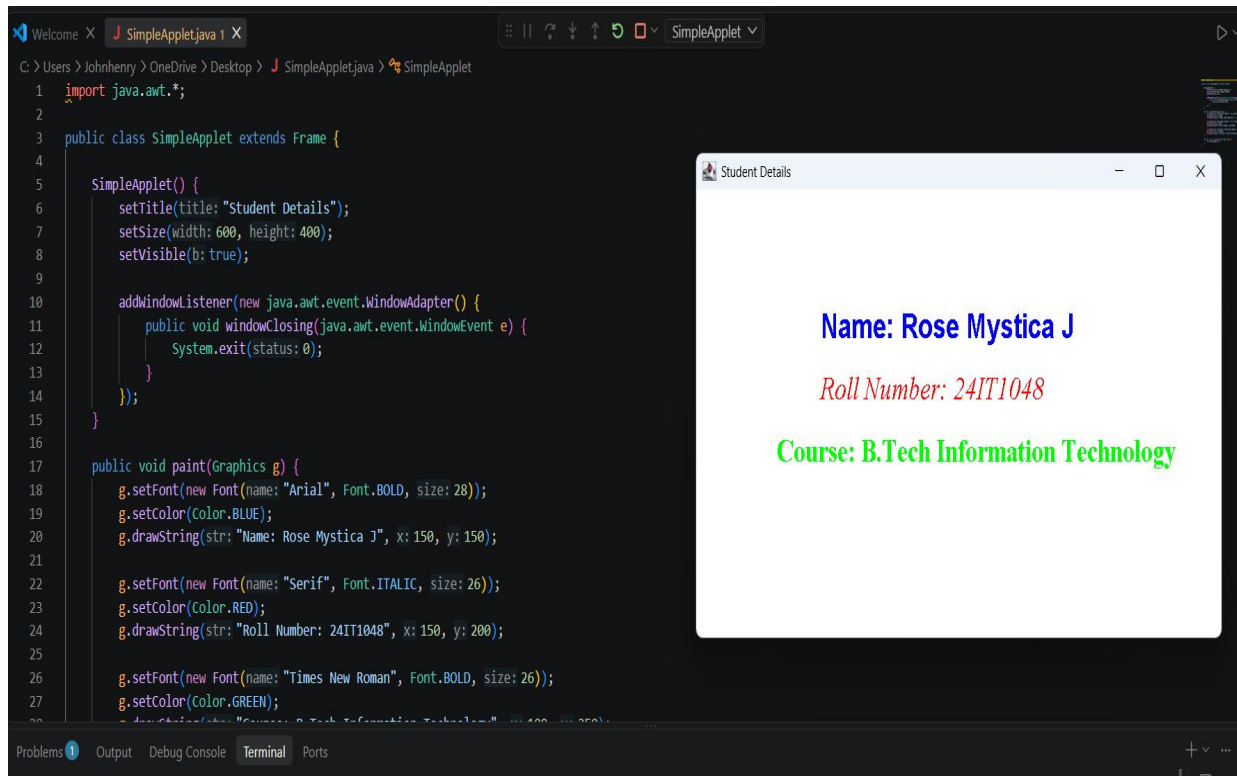
The screenshot displays an IDE with a Java file named `BasicShapes.java`. The code defines a class `BasicShapes` that extends `Frame` and implements `ItemListener`. It includes two `Choice` controls: `shape` and `color`. The `shape` control has items "Line", "Rectangle", and "Oval". The `color` control has items "Red", "Blue", "Green", and "Black". The `BasicShapes()` constructor sets the title to "Basic Shapes", size to 600x400, and layout to `FlowLayout`. It adds the `shape` and `color` controls and a label "Select Shape:". The application is running, and the window titled "Basic Shapes" is visible. It shows the "Select Shape:" label, a "Select Color:" label, and a red rectangle. The "Select Shape:" dropdown is set to "Rectangle" and the "Select Color:" dropdown is set to "Red".

```
1 import java.awt.*;
2 import java.awt.event.*;
3
4 public class BasicShapes extends Frame implements ItemListener {
5
6     Choice shape, color;
7     String s = "Line";
8     Color c = Color.RED;
9
10    BasicShapes() {
11
12        setTitle(title: "Basic Shapes");
13        setSize(width: 600, height: 400);
14        setLayout(new FlowLayout());
15
16        shape = new Choice();
17        shape.add(item: "Line");
18        shape.add(item: "Rectangle");
19        shape.add(item: "Oval");
20
21        color = new Choice();
22        color.add(item: "Red");
23        color.add(item: "Blue");
24        color.add(item: "Green");
25        color.add(item: "Black");
26
27        add(new Label(text: "Select Shape:"));
28    }
```

Q2 a simple Applet

25 marks

Develop a simple Applet that displays your name, roll number, and course in different colors and fonts. Use the paint() method for rendering.



The screenshot shows an IDE with a Java file named `SimpleApplet.java`. The code defines a class `SimpleApplet` that extends `Frame`. It sets the title to "Student Details", size to 600x400, and is visible. It adds a window listener that calls `System.exit(0)` when the window is closed. The `paint` method uses `Graphics` to draw the student's name, roll number, and course in different fonts and colors.

```
1 import java.awt.*;
2
3 public class SimpleApplet extends Frame {
4
5     SimpleApplet() {
6         setTitle(title: "Student Details");
7         setSize(width: 600, height: 400);
8         setVisible(b: true);
9
10        addWindowListener(new java.awt.event.WindowAdapter() {
11            public void windowClosing(java.awt.event.WindowEvent e) {
12                System.exit(status: 0);
13            }
14        });
15    }
16
17    public void paint(Graphics g) {
18        g.setFont(new Font(name: "Arial", Font.BOLD, size: 28));
19        g.setColor(color: BLUE);
20        g.drawString(str: "Name: Rose Mystica J", x: 150, y: 150);
21
22        g.setFont(new Font(name: "Serif", Font.ITALIC, size: 26));
23        g.setColor(color: RED);
24        g.drawString(str: "Roll Number: 24IT1048", x: 150, y: 200);
25
26        g.setFont(new Font(name: "Times New Roman", Font.BOLD, size: 26));
27        g.setColor(color: GREEN);
28        g.drawString(str: "Course: B.Tech Information Technology", x: 150, y: 250);
29    }
30 }
```

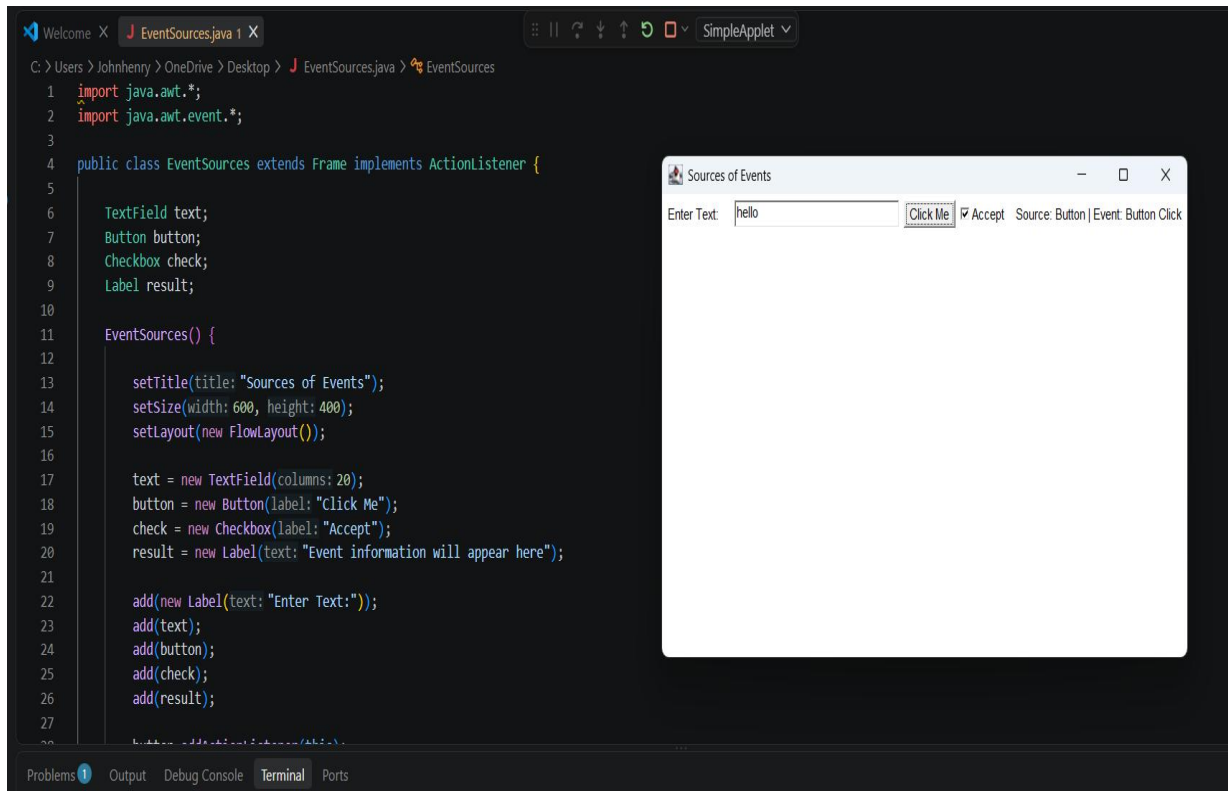
The rendered output is a window titled "Student Details" with the following text:

Name: Rose Mystica J
Roll Number: 24IT1048
Course: B.Tech Information Technology

Q3 sources of events

25 marks

Write a program that demonstrates different sources of events (Button click, TextField input, Checkbox selection, Window closing). Display the event source and type



The screenshot shows an IDE with a Java file named `EventSources.java` and a running application window titled "Sources of Events".

Java Code:

```
1 import java.awt.*;
2 import java.awt.event.*;
3
4 public class EventSources extends Frame implements ActionListener {
5
6     TextField text;
7     Button button;
8     Checkbox check;
9     Label result;
10
11     EventSources() {
12
13         setTitle(title: "Sources of Events");
14         setSize(width: 600, height: 400);
15         setLayout(new FlowLayout());
16
17         text = new TextField(columns: 20);
18         button = new Button(label: "Click Me");
19         check = new Checkbox(label: "Accept");
20         result = new Label(text: "Event information will appear here");
21
22         add(new Label(text: "Enter Text:"));
23         add(text);
24         add(button);
25         add(check);
26         add(result);
27     }
28 }
```

Running Application:

The application window "Sources of Events" displays the following UI elements:

- A label "Enter Text:" followed by a text field containing the text "hello".
- A button labeled "Click Me".
- A checkbox labeled "Accept" which is currently checked.
- A label at the bottom right that reads "Source: Button | Event: Button Click".

Q4 menu driven

1 mark

Create a menu-driven Swing application for a simple calculator (Add, Subtract, Multiply, Divide, Clear, Exit) using MenuBar, Menu, and MenuItem.

The screenshot displays an IDE with the following components:

- Code Editor:** Shows the `MenuCalculator.java` file. The code defines a `MenuCalculator` class extending `JFrame`. It includes imports for `javax.swing`, `java.awt`, and `java.awt.event`. The `MenuCalculator()` constructor sets the window title to "Menu Driven Calculator", size to 550x400, and layout to null. It adds a title label "MENU DRIVEN CALCULATOR" with a bold font. It also adds a label "Number 1:" and a text field for the first number.
- Application Window:** A window titled "Menu Driven Calculator" is shown. It features a menu bar with the following items: Add, Subtract, Multiply, Divide, Clear, and Exit. The main area of the window displays the text "MENU DRIVEN CALCULATOR". Below this, there are two input fields labeled "Number 1:" and "Number 2:". The "Number 1:" field contains the value 3, and the "Number 2:" field contains the value 5. Below these fields, there is a label "Result:" followed by a text field containing the value 8.0.